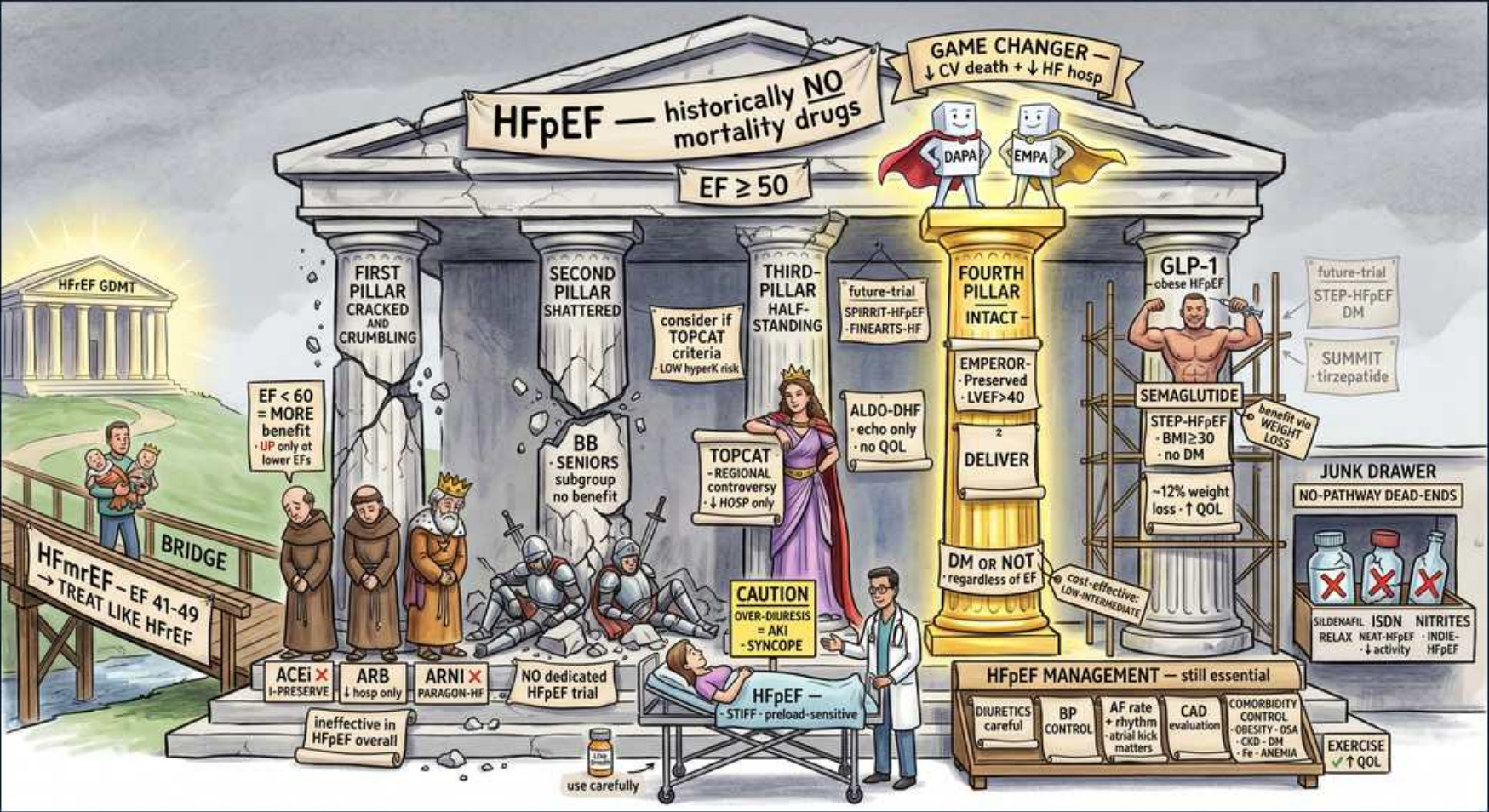


Heart Failure — HFpEF Management & Comorbidities





## 1. HFpEF — EF ≥ 50

### WHAT YOU SEE

Temple title HFpEF, EF >= 50

### WHAT IT ENCODES

HFpEF = HF with preserved EF (LVEF ≥50%) plus structural/functional evidence of raised filling pressures (LV hypertrophy, LA enlargement, elevated E/e', raised BNP). It is a disease of a stiff, non-compliant ventricle with diastolic dysfunction, driven by HTN, obesity, DM, AF and aging. Historically NO drug reduced mortality — the entire treatment paradigm is congestion control plus the SGLT2i breakthrough. Use the H2FPEF or HFA-PEFF score to make the diagnosis when EF is normal but dyspnea is unexplained.

### BOARD TRAP

WRONG: HFpEF and HFrEF share the same proven four mortality-reducing pillars. ARNI, ACEi/ARB and beta-blockers have NOT shown clear mortality benefit in HFpEF; only SGLT2i (and to a borderline degree MRA) have evidence. The discriminator: HFrEF GDMT is reflexive, HFpEF treatment is congestion + comorbidity control + SGLT2i.

## 2. HFmrEF — EF 41-49

### WHAT YOU SEE

Bridge label HFmrEF, treat like HFrEF

### WHAT IT ENCODES

HFmrEF (mildly reduced EF, LVEF 41–49%) is the bridge category. Guidelines say treat like HFrEF — give the four pillars (ARNI/ACEi, beta-blocker, MRA, SGLT2i) even though the evidence is weaker and largely extrapolated/subgroup-derived. Many HFmrEF patients are recovering HFrEF or trending in either direction, so serial echo matters.

### BOARD TRAP

WRONG: HFmrEF gets no GDMT because EF is 'almost normal.' Discriminator: the 2022 AHA/ACC/HFSA guideline gives a IIb–IIa recommendation to use the HFrEF pillars in HFmrEF; SGLT2i has the strongest support across the EF spectrum.

## 3. ACEi/ARB/ARNI — weak in HFpEF

### WHAT YOU SEE

First pillar cracked, ACEi/ARB X, ARNI partial

### WHAT IT ENCODES

In HFpEF the RAAS pillar is cracked: ACEi (PEP-CHF/perindopril) and ARB (CHARM-Preserved candesartan, I-PRESERVE irbesartan) failed to reduce mortality. ARNI sacubitril-valsartan in PARAGON-HF narrowly MISSED its primary endpoint (p=0.06) with benefit only in the lower-EF and female subgroups. Net: no class-wide HFpEF mortality benefit.

### BOARD TRAP

WRONG: ARNI clearly reduces HFpEF mortality. Discriminator: PARAGON-HF was statistically negative; ARNI is only FDA-labeled with a soft 'may be considered' nod in HFpEF, strongest where EF is below normal.

## 4. Beta-blockers — no HFpEF benefit

### WHAT YOU SEE

Second pillar shattered, BB no benefit

### WHAT IT ENCODES

Beta-blockers have NO proven mortality or hospitalization benefit in HFpEF — there is no dedicated positive trial. They can be harmful by blunting the chronotropic response, because a stiff ventricle is preload/rate dependent and relies on heart rate to maintain output. Only use a beta-blocker in HFpEF for a separate indication (AF rate control, angina, prior MI).

### BOARD TRAP

WRONG: every HF patient should be on a beta-blocker. Discriminator: beta-blocker mortality benefit is a HFrEF/HFmrEF phenomenon; in pure HFpEF excessive bradycardia worsens diastolic filling and exercise tolerance.

## 5. MRA — TOPCAT borderline

### WHAT YOU SEE

Third pillar half-standing, TOPCAT, aldo

### WHAT IT ENCODES

MRA (spironolactone 25–50 mg) in TOPCAT had a borderline/regional result — neutral overall but reduced HF hospitalization, with apparent benefit in the Americas (the Russia/Georgia sites had suspect drug-level data). Net: MRA may reduce HF hospitalization in HFpEF, especially at the lower end of the EF range, while you watch K+ and creatinine.

### BOARD TRAP

WRONG: TOPCAT proved spironolactone reduces HFpEF mortality everywhere. Discriminator: the overall trial was neutral; the 'benefit' is a post-hoc regional/secondary-endpoint signal, so MRA is a IIb consideration, not a mandate.

## 6. SGLT2i — the game changer

### WHAT YOU SEE

Dapa/Empa crown, game changer banner

### WHAT IT ENCODES

SGLT2i are THE game changer in HFpEF: dapagliflozin (DELIVER) and empagliflozin (EMPEROR-Preserved) are the first drugs to reduce the composite of CV death + HF hospitalization across the full EF spectrum, regardless of diabetes. Fixed dose 10 mg once daily, no titration. Mechanism is osmotic/natriuretic diuresis plus favorable cardiac-metabolic and anti-fibrotic effects. Watch for euglycemic DKA and genital mycotic infections.

### BOARD TRAP

WRONG: no drug helps HFpEF / SGLT2i only works in diabetics. Discriminator: DELIVER and EMPEROR-Preserved enrolled non-diabetics and still showed benefit — SGLT2i is now Class I for HFpEF independent of glycemic status.

## 7. Fourth pillar intact — SGLT2i

### WHAT YOU SEE

Intact pillar, EMPEROR-Preserved/DELIVER

### WHAT IT ENCODES

SGLT2i is the ONLY fully intact pillar across the HF spectrum — it works at LVEF >40% (EMPEROR-Preserved, DELIVER) and at LVEF ≤40% (EMPEROR-Reduced, DAPA-HF), with or without diabetes and with or without CKD. This is why it is the unifying first move when you confirm any phenotype of HF.

### BOARD TRAP

WRONG: hold SGLT2i because the patient isn't diabetic or has CKD with eGFR 25. Discriminator: benefit is independent of DM, and SGLT2i are now started down to eGFR ~20–25; the expected small initial creatinine bump is hemodynamic, not a reason to stop.



## 8. GLP-1 RA / semaglutide

### WHAT YOU SEE

GLP-1 strongman, STEP-HFpEF

### WHAT IT ENCODES

Semaglutide (STEP-HFpEF / STEP-HFpEF DM) in OBESE HFpEF improves symptoms (KCCQ score), exercise capacity (6-minute walk), weight and inflammation (CRP). It targets the obesity-driven inflammatory HFpEF phenotype. Note this is symptom/QOL data, not a proven mortality reduction.

### BOARD TRAP

WRONG: semaglutide is a proven mortality-reducing HFpEF drug. Discriminator: STEP-HFpEF measured symptoms/function/weight, not mortality — give it for the obese HFpEF phenotype, not as a substitute for SGLT2i.

## 9. BP control

### WHAT YOU SEE

The 'BP CONTROL' block in the 'HFpEF MANAGEMENT — still essential' base frieze — controlling the hypertension that drives the LVH and diastolic stiffness.

### WHAT IT ENCODES

Aggressive BP control is the foundation of HFpEF management because HTN is the main driver of the LV hypertrophy and diastolic stiffness. Target roughly <130/80. Prefer agents that also help HF/comorbidity (ARB, MRA, with diuretic for volume); SPRINT-type intensive control reduces incident HF.

### BOARD TRAP

WRONG: lenient BP targets are fine in HFpEF since EF is normal. Discriminator: uncontrolled HTN directly worsens diastolic filling pressures and precipitates flash pulmonary edema — BP control is disease-modifying here.

## 10. AF management

### WHAT YOU SEE

The 'AF — rate v rhythm — atrial kick matters' block in the HFpEF-management base frieze — losing the atrial kick is costly in a stiff ventricle.

### WHAT IT ENCODES

AF is poorly tolerated in HFpEF because loss of the atrial kick removes up to ~20% of diastolic filling in a stiff ventricle, and rapid rates shorten the already-critical filling time. Control rate, and consider rhythm control/ablation since restoring sinus rhythm can markedly improve symptoms. Anticoagulate per CHA2DS2-VASc.

### BOARD TRAP

WRONG: rate vs rhythm doesn't matter in HFpEF as long as it's anticoagulated. Discriminator: because diastolic filling depends so heavily on the atrial kick and a slow rate, maintaining sinus rhythm is often more symptomatically valuable in HFpEF than in HFrEF.

## 11. Diuretics — careful

### WHAT YOU SEE

The 'DIURETICS — careful' block in the HFpEF-management base frieze, captioned 'use carefully' — the stiff, preload-sensitive ventricle punishes over-diuresis.

### WHAT IT ENCODES

Loop diuretics relieve congestion in HFpEF, but use them carefully: the stiff ventricle sits on the steep part of the pressure–volume curve and is exquisitely preload-sensitive, so over-diuresis drops cardiac output, causes hypotension and pre-renal AKI. Titrate to euvolemia, not to a dry target.

### BOARD TRAP

WRONG: HFpEF tolerates aggressive diuresis as well as HFrEF. Discriminator: in HFpEF small volume shifts cause big pressure/output swings — over-diuresis triggers hypotension and azotemia, so you diurese gently.

## 12. Junk drawer — no benefit

### WHAT YOU SEE

The 'JUNK DRAWER — NO-PATHWAY DEAD-ENDS' shelf of red-X'd bottles (digoxin, ISDN/nitrates with 'NEAT-HFpEF ↓ activity', sildenafil 'RELAX', inhaled nitrite 'INDIE-HFpEF') — the HFpEF therapies proven useless or harmful.

### WHAT IT ENCODES

HFpEF dead-ends with no benefit (or harm): digoxin (no HFpEF data), and nitrates/isosorbide dinitrate — NEAT-HFpEF actually showed LESS daily activity with isosorbide mononitrate, and INDIE-HFpEF showed no benefit from inhaled nitrite. Phosphodiesterase-5 inhibitors (RELAX, sildenafil) were also negative.

### BOARD TRAP

WRONG: nitrates improve exercise capacity in HFpEF. Discriminator: NEAT-HFpEF found nitrates REDUCED activity levels — avoid routine nitrate/PDE5i therapy in HFpEF; they belong in the junk drawer.

## 13. Iron deficiency — IV iron

### WHAT YOU SEE

The iron-deficiency / anemia comorbidity callout — diagnose by ferritin <100 (or 100–299 with TSAT <20%) and treat with IV ferric carboxymaltose, never oral (IRONOUT-HF).

### WHAT IT ENCODES

Iron deficiency is present in up to ~50% of HF patients and worsens symptoms and exercise capacity even WITHOUT anemia. Diagnose by ferritin <100, or ferritin 100–299 with transferrin saturation <20%. Treat symptomatic HFrEF (and increasingly HFmrEF/HFpEF) with IV iron — ferric carboxymaltose (AFFIRM-AHF, FAIR-HF, CONFIRM-HF) improves symptoms/QOL and reduces HF hospitalizations.

### BOARD TRAP

WRONG: give ORAL iron to correct iron deficiency in HF. Discriminator: IRONOUT-HF showed oral iron does NOT replete iron stores or improve outcomes in HF — the answer is IV ferric carboxymaltose, and you treat the deficiency even when hemoglobin is normal.

## 14. Sleep apnea — treat OSA, caution CSA

### WHAT YOU SEE

The sleep-disordered-breathing comorbidity callout — CPAP for obstructive apnea, but adaptive servo-ventilation is CONTRAINDICATED in HFrEF with central apnea (SERVE-HF).

### WHAT IT ENCODES

Sleep-disordered breathing is common in HF. Obstructive sleep apnea worsens afterload and arrhythmia — treat with CPAP. Central sleep apnea (Cheyne-Stokes) marks advanced HF; treat the HF itself. Critically, adaptive servo-ventilation (ASV) is CONTRAINDICATED in HFrEF with predominant central sleep apnea.

### BOARD TRAP

WRONG: use adaptive servo-ventilation (ASV) for central sleep apnea in HFrEF. Discriminator: the SERVE-HF trial showed ASV INCREASED cardiovascular mortality in HFrEF (EF ≤45%) with central sleep apnea — ASV is contraindicated there; CPAP is for OSA, not CSA in HFrEF.



## 15. Depression / AF comorbidity

### WHAT YOU SEE

The depression + AF-burden comorbidity callout — screen/treat depression (SSRI, not TCA) and rate/rhythm-control plus anticoagulate the AF that both causes and worsens HF.

### WHAT IT ENCODES

Depression affects ~20–40% of HF patients and independently predicts worse adherence, more hospitalizations and higher mortality — screen and treat. Atrial fibrillation both causes and complicates HF (tachycardia-mediated cardiomyopathy); rate/rhythm control plus anticoagulation per CHA<sub>2</sub>DS<sub>2</sub>-VASc, and catheter ablation (CASTLE-AF) reduces mortality/hospitalization in selected HFrEF + AF patients.

### BOARD TRAP

WRONG: treat HF depression with a tricyclic antidepressant. Discriminator: TCAs are pro-arrhythmic and cause orthostasis — avoid them in HF; SSRIs are preferred (though SADHART-CHF/MOOD-HF show modest efficacy), and never ignore depression as it drives non-adherence and readmission.

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